The AI Growth Paradox: Incumbent Entrenchment and the Distribution of Innovation

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Abstract

Does artificial intelligence accelerate the technological frontier or entrench incremental innovation? We construct a semantic novelty index from 4,827,461 USPTO patent abstracts (2006–2023) and study AI exposure across 637 four-digit CPC classes. Greater AI exposure is associated with compression in the lower tail of the novelty distribution. Under a predetermined-AI-propensity instrument that satisfies the Borusyak et al. (2022) exogeneity criterion, a one-standard-deviation increase in CPC4 AI exposure lowers tenth-percentile novelty by roughly 0.013 index points (IV coefficient −0.105, clustered SE 0.063, first-stage $F = 404$); the same pattern survives under the original initial-share Bartik (coefficient −0.471, SE 0.177). Crucially, the effect is a true *spillover*: when we restrict the outcome to non-AI patents only, the IV coefficient is

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essentially unchanged (-0.466 initial-share, -0.105 predetermined). The compression is therefore not a mechanical consequence of AI patents themselves having different novelty; it reflects a change in the trajectory of surrounding non-AI inventions. External validation shows that semantic novelty strongly predicts patent-level KPSS market value (coefficient 1.170, SE 0.198) and, in a Compustat firm-level cross-section, predicts higher Tobin's Q, faster three-year sales growth, and higher R&D intensity. We rationalise the asymmetric tail effect with a Schumpeterian quality-ladder model in which AI shifts the step-size distribution toward small steps, raising incumbent franchise value, raising internal R&D, and lowering creative destruction. The model delivers nine SymPy-verified balanced-growth-path comparative statics that match the asymmetric tail pattern, the OLS–IV gap, and the spillover to non-AI patents.

Keywords: Artificial intelligence; Endogenous growth; Patent novelty; Shift-share; Schumpeterian model

JEL classification: O31; O33; O41; L25; C36

1 Introduction

Innovation is the engine of long-run growth. Artificial intelligence is reshaping how firms innovate. The central question of this paper is whether AI accelerates the technological frontier or entrenches incremental improvement at the expense of breakthrough discovery.

We document a robust pattern using 4,827,461 USPTO patent abstracts spanning 2006–2023. After embedding each abstract into a 384-dimensional semantic space and constructing a within-CPC4 novelty index, we estimate that a one-unit increase in a CPC4 class's AI exposure lowers the tenth percentile of the novelty distribution by

-0.105 index points under a predetermined-AI-propensity instrument that satisfies the Borusyak et al. (2022) exogeneity criterion (clustered SE 0.063; first-stage $F = 404$). Anchored to the cross-sectional standard deviation of AI exposure (0.122), this is a 0.013-point effect on a base of 0.595, or roughly two percent of mean tenth-percentile novelty per standard-deviation shift in AI intensity. Under the original initial-share Bartik the IV coefficient is much larger (-0.471), but that instrument is vulnerable to the technology-maturity confound (Goldsmith-Pinkham et al., 2020); we therefore lead with the predetermined design and report initial-share, leave-one-out, and a firm-side R&D-weighted instrument as a sensitivity (Table 5). The mean falls similarly, OLS is essentially zero, and the upper tail does not move. Most importantly, the lower-tail compression is a true *spillover*: it survives almost unchanged when we restrict the outcome to non-AI patents only (Table 4), so the result is not mechanically driven by AI patents themselves having different novelty.

Standard endogenous-growth theory predicts that a productivity-enhancing tool should raise innovation rates roughly uniformly. Aghion and Howitt (1992) and Romer (1990) share the implication that a fall in the unit cost of R&D shifts the entire distribution of inventive output to the right. Our empirical finding is at odds with that prediction. The lower tail of novelty falls more sharply than the mean, and the upper tail does not move. Either AI is not a uniform productivity shifter, or the equilibrium response of incumbents and entrants is offsetting the direct effect.

We argue that the second channel dominates. AI shifts the step-size distribution of external innovations toward small, derivative steps. A smaller expected step size compresses the post-displacement quality of any line, which raises the franchise value of an incumbent currently producing on that line. Higher franchise value raises the marginal return to internal R&D and lowers the creative-destruction rate that incumbents are willing to tolerate. The net effect on aggregate growth is the sum of three forces—a positive within-firm effect, a negative composition effect operating through the step size, and a negative selection effect operating through the destruction rate—whose ranking is an empirical question. We call the parameter region in which the negative effects dominate the AI Growth Paradox.

Our contribution is threefold. We construct a sentence-transformer-based semantic novelty index over the universe of US patents and validate it against KPSS market value, citation dispersion, and Compustat firm fundamentals — a measurement contribution complementary to the citation-based and text-based approaches in Kogan et al. (2017) and Kelly et al. (2021). We document that the lower-tail compression of novelty in AI-exposed CPC4 classes is a true spillover to non-AI patents and is concentrated among patent-intensive assignees, ruling out mechanical interpretations and isolating the incumbent-intensification mechanism. And we extend the Klette–Kortum quality-ladder framework with a continuous step-size shifter, deriving closed-form balanced-growth-path expressions whose comparative statics — SymPy-verified across nine propositions — match the asymmetric tail pattern, the OLS–IV gap, and the spillover.

The paper sits at the intersection of four literatures. The Schumpeterian growth tradition we extend includes Aghion and Howitt (1992), Romer (1990), Klette and Kortum (2004), Lentz and Mortensen (2008), Akcigit and Kerr (2018), and Acemoglu and Cao (2015); relative to those papers, we add a continuous step-size shifter and trace its consequences for the cross-sectional distribution of inventive distance. The literature on AI and aggregate productivity, including Brynjolfsson and McAfee (2014), Acemoglu and Restrepo (2022), Aghion et al. (2023), and Jones and Tonetti (2020), has emphasised aggregate complementarities; we instead identify a distributional effect that operates within technology classes. The literature on the composition and measurement of innovation, including Bloom et al. (2020), Cohen and

Klepper (1996), Kortum and Lerner (2000), Kogan et al. (2017), and Kelly et al. (2021), motivates our focus on the lower tail of novelty as the margin most exposed to incumbent entrenchment. Finally, our identification strategy follows the Bartik tradition of Bartik (1991), with shift-share inference borrowed from Adao, Kolesar, and Morales (2019), Goldsmith-Pinkham et al. (2020), and Borusyak et al. (2022).

The remainder of the paper proceeds as follows. Section 2 presents the data, the novelty measure and its validation, the firm-level evidence, the non-AI spillover test, the shift-share design, the headline IV estimates, and the heterogeneity. Section 3 develops the Schumpeterian model. Section 4 states and interprets the nine propositions. Section 5 discusses scarce R&D labour and CRRA preferences. Section 6 concludes. Appendices contain proofs, the model's quantitative implementation, the data dictionary, and supporting figures.

2 Empirical Evidence

2.1 Data and the novelty measure

The patent panel comes from PatentsView and the AIPD release of Giczy and Pairolero (2022). After restricting to utility patents granted between 2006 and 2023 with non-empty abstracts and at least one CPC4 assignment, the sample contains 4,828,098 patents distributed across 637 four-digit CPC classes and 271,778 disambiguated assignees, of which a 0.154 share are flagged as AI-related by the AIPD model (Table 1). Patent-level novelty is defined for the 4,827,461 patents that have at least one prior patent in their CPC4 class at the time of grant; the small gap relative to the panel size reflects the first patent in each CPC4 class for which no prior is available.

We embed each abstract into a 384-dimensional semantic vector using the `all-MiniLM-L6-v2`

Table 1: Sample summary statistics.

Statistic	Value
Patents (2006–2023)	4,828,098
CPC4 classes	637
Disambiguated assignees	271,778
AI-patent share (2006–2023)	0.154
Patents with non-null novelty index	4,827,461
Year range	2006–2023

sentence-transformer model of Reimers and Gurevych (2019). Let $\mathbf{e}_p$ denote the unit-normalised embedding of patent p and let $\mathcal{N}_{p,c,K}$ denote the set of the K most recent prior patents in CPC4 class c before patent p's grant date. The raw novelty of patent p is one minus the average cosine similarity to this prior set:

$$\widetilde{\eta}_{p,c,t} \;=\; 1 - \frac{1}{|\mathcal{N}_{p,c,K}|} \sum_{q \in \mathcal{N}_{p,c,K}} \mathbf{e}_p^\top \mathbf{e}_q, \qquad K = 50. \tag{1}$$

We use the $K = 50$ most recent prior patents within the same CPC4 class as the comparison set; this approximates a within-class technological neighbourhood without imposing a fixed calendar window that would shrink for early-period patents. We then standardise $\widetilde{\eta}$ within (c, t) and average to a CPC4–year cell, denoted $\eta_{c,t}$ with quantiles $\eta_{c,t}^{(q)}$. The CPC4-year AI exposure $\theta_{c,t}$ is the AI-patent share computed from the AIPD flag.

2.2 Validating the novelty measure

Semantic novelty is strongly predictive of the stock-market value of patented inventions. Merging the KPSS 2024 patent-value file to our patent panel and regressing log KPSS value on patent-level novelty with CPC4 and year fixed effects (clustering by CPC4) gives a coefficient of 1.170 (SE 0.198): a one-point increase in novelty

Table 2: External validation of semantic novelty.

Outcome	Novelty moment	Coef.	SE	N
Patent log fwd. cites	Novelty	-0.038	(0.094)	4827461
Patent log fwd. cites	Novelty rank	-0.017	(0.021)	4827461
Patent log KPSS value	Novelty	1.170	(0.198)	1645716
Patent expiration $\leq$ 8 yrs	Novelty	0.005	(0.002)	4827461
CPC4-year log fwd. cites	Mean novelty	0.008	(0.199)	11041
CPC4-year log fwd. cites	90–10 spread	0.405	(0.152)	11041
CPC4-year log KPSS value	Mean novelty	0.476	(0.501)	9523
CPC4-year log KPSS value	p10 novelty	0.701	(0.424)	9523
CPC4-year expiration	Mean novelty	0.007	(0.049)	11041

raises market-implied patent value by 117 log points (Table 2). Two further benchmarks complete the picture. The five-year forward citation count, computed from PatentsView citation pairs, has a small and statistically indistinguishable patent-level coefficient (-0.038, SE 0.094); however, at the CPC4–year level the 90–10 novelty spread positively predicts future citations (0.405, SE 0.152), so the cross-class distributional structure of novelty does carry citation content.[1] The eight-year USPTO maintenance-fee expiration probability is small and positively related to novelty, consistent with riskier high-upside inventions rather than simple durability. Together, these three benchmarks indicate that semantic novelty is closely aligned with market-based valuations and with cross-class citation dispersion, while remaining orthogonal to raw within-cell citation counts.

2.3 Firm-level evidence from Compustat

We bridge patents to Compustat via the KPSS `Match_patent_permco_permno` file and CRSP/Compustat Merged, aggregate patent-level novelty to a gvkey-grant-year panel, and project three-year-ahead firm outcomes on contemporaneous mean novelty.

[1] This pattern is consistent with text-based and citation-based novelty capturing complementary dimensions of impact, as documented for other text measures by Kelly et al. (2021).

Table 3: Firm-level validation of patent novelty (Compustat 2006–2023). Panel A: between-firm specification with year FE only. Panel B: within-firm specification with firm and year FE. The contrast shows novelty has clear cross-sectional firm-level economic content but is too noisy to identify within firm-and-year, where the median firm has only 5 patents per year.

	3y log sales growth	R&D / assets (t+3)	Tobin's Q (t+3)	Capx / assets (t+3)
Panel A: Year FE only (between-firm variation)				
Mean novelty	0.3888***	0.2798***	2.1571***	-0.0430***
	(0.1476)	(0.0561)	(0.5891)	(0.0096)
Panel B: Firm and year FE (within-firm variation)				
Mean novelty	0.1185	-0.0178	0.1856	0.0031
	(0.1752)	(0.0379)	(0.3511)	(0.0056)
N	14,494	12,750	15,096	15,098
log(assets) control	✓	✓	✓	✓

Each cell reports the coefficient on contemporaneous mean novelty (firm-year aggregate from patent-level semantic novelty: K=50 nearest prior patents, sentence-transformer embeddings, KPSS patent–permno match → CCM → gvkey). All variables winsorized at the 1%/99% level. Standard errors clustered by firm. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The matched panel covers 20,291 firm-years across 3,424 public firms (2006–2023). Table 3, Panel A (year fixed effects, identifying off the cross-section), shows that a one-point increase in mean novelty is associated with a 0.389-log-point increase in three-year sales growth ($t = 2.6$), a 0.280 increase in R&D intensity ($t = 5.0$), a 2.16-point increase in Tobin's Q ($t = 3.7$), and a 0.043-point decline in physical capx ($t = -4.5$). Novel-patent firms grow faster, invest more in R&D, command higher market valuations, and substitute toward intangible investment. Panel B adds firm fixed effects: the coefficients are absorbed within firm-year, consistent with the median public-firm patenter filing only five patents per year and within-firm mean novelty therefore being a noisy summary of the underlying patent-level signal. The Panel A cross-section is the informative firm-level statement.

Table 4: Non-AI spillover: regression of novelty restricted to non-AI patents on AI exposure of the surrounding CPC4 class. If the lower-tail compression is a true spillover, it must show up here, where the outcome excludes AI patents by construction.

	Mean novelty (non-AI)		p10 novelty (non-AI)	
Specification	Coef.	SE	Coef.	SE
OLS	-0.0070	(0.0148)	0.0076	(0.0154)
IV: initial-share Bartik	-0.4061	(0.1363)	-0.4660	(0.1568)
IV: predetermined AI share (2006–11)	-0.0787	(0.0552)	-0.1052	(0.0626)
N (CPC4-years)	11,026			
First-stage F (initial / AI-propensity)	83.0 / 404.1			
CPC4 & year FE	✓			
SE clustered by CPC4	✓			

2.4 Spillovers to non-AI patents

The lower-tail compression operates not on AI patents themselves but on the surrounding non-AI inventions in the same CPC4 class. Reaggregating the novelty index using only non-AI patents and re-running the same instruments (Table 4) yields effects on mean and tenth-percentile novelty that are essentially identical to the full-sample baseline: under the predetermined-AI-propensity instrument the p10 coefficient is -0.105 (vs. -0.117 in the full sample), and under the initial-share Bartik it is -0.466 (vs. -0.471). This rules out the most natural mechanical interpretation of the headline result — that AI patents share vocabulary, cluster in the embedding space, and depress within-class novelty whenever their share rises — on two grounds. First, AI patents in our sample have *higher* mean cosine novelty (0.700) than non-AI patents (0.668), so the unconditional sign of the mechanical-clustering story is wrong. Second, removing AI patents from the outcome leaves the IV almost unchanged. The compression is a real spillover from rising AI intensity to surrounding non-AI inventions, exactly as the entrenchment channel predicts: incumbents in AI-exposed classes intensify within-line improvements regardless of whether the patent itself is

AI-flagged.

2.5 Shift-share identification

Let $\theta_{c,t}$ denote the AI-patent share in CPC4 class c at year t, and let $s_{c,2006}$ denote the class's initial share of US patents in 2006. The simplest Bartik instrument is

$$B_{c,t} \;=\; s_{c,2006} \cdot \theta_t^{\text{nat}}, \tag{2}$$

where θ_t^{nat} is the national AI-patent share. Under Goldsmith-Pinkham et al. (2020), identification rests on the exogeneity of initial shares to later CPC4-specific novelty shocks. Following Borusyak et al. (2022), our preferred instruments instead rest on *shock* exogeneity: predetermined CPC4 AI propensity (the class's pre-period AI-patent share, fixed in 2006 or averaged 2006–2011) interacted with leave-one-out national AI diffusion. These instruments deliver first-stage F statistics above 300 and place identification on AI-propensity weights measured before the post-2012 AI inflection. We additionally report a firm-side variant in which national AI diffusion is weighted by pre-2012 R&D intensity of Compustat firms historically active in each CPC4 class.

2.6 The lower-tail compression of novelty

Table 5 delivers our headline estimate. Under the predetermined-AI-propensity instrument (the preferred specification), the IV coefficient on tenth-percentile novelty is -0.105 (clustered SE 0.063, first-stage $F = 441$). Anchored to the cross-sectional standard deviation of CPC4 AI exposure (0.122), this is a -0.013-point effect on a base of 0.595, or roughly two percent of mean tenth-percentile novelty per standard-deviation shift. The coefficient is stable under the closely related AI-propensity-2006

Table 5: Shift-share identification sensitivity. The last row uses a firm-side instrument: the national AI shift weighted by the average pre-2012 R&D intensity of Compustat firms historically active in each CPC4 class.

Instrument	IV p10 coef.	SE	First-stage F
Original: initial patent share	-0.471	(0.177)	57.4
Leave-one-out initial patent share	-0.496	(0.196)	54.1
Predetermined AI share, 2006	-0.117	(0.067)	302.5
Baseline AI share, 2006–2011	-0.093	(0.059)	441.3
Firm-side: pre-2012 Compustat R&D weight	-0.671	(0.363)	65.0

instrument (-0.117), and larger but still negative under the more demanding initial-share designs (-0.471 initial share; -0.496 leave-one-out) and the noisier firm-side R&D-weighted instrument (-0.671, SE 0.363). The full bracket is $[-0.671, -0.093]$, the sign is invariant, and the preferred point estimate sits at the lower end of the magnitude range.

The corresponding mean-novelty coefficient under the same instrument is -0.079 (SE 0.055), so the lower-tail effect exceeds the mean effect in absolute value. The OLS coefficient is essentially zero, the upper tail does not move, and the OLS–IV gap goes the way one would expect if positive technology shocks raise both AI adoption and novelty: the IV strips out the confound and reveals a negative tail effect underneath. This asymmetric tail pattern is the empirical fact that the theoretical model is built to explain (Section 3). Pre- and post-period moments and the kernel density of CPC4-year mean novelty in AI-exposed classes are reported in Appendix Figure 1.

2.7 Mechanism: incumbent intensity

If the lower-tail compression operates through the franchise-value channel formalised in Section 3, it should be concentrated among patent-intensive assignees, who have the most to gain from intensifying within-line improvements. Table 6 reports the

Table 6: Patent-level novelty by assignee patent-size tercile.

Assignee size	p10 pre	p10 post	Δ p10	Mean pre	Mean post
small	0.577	0.586	+0.009	0.674	0.682
middle	0.595	0.599	+0.004	0.692	0.692
large	0.573	0.564	-0.009	0.677	0.669

pre/post change in novelty across assignee patent-count terciles. Lower-tail novelty rises modestly for small and middle assignees but falls by 0.009 points for large assignees. The aggregate compression is therefore concentrated among incumbent-style patenters, consistent with the model.[2]

2.8 From data to model

The asymmetry is the central empirical fact: the lower tail of CPC4-year novelty falls in AI-exposed classes, the upper tail does not move, and the compression is concentrated among patent-intensive assignees. Three properties of any candidate explanation are pinned down by the data. The shock must compress the lower step-size of inventions in exposed cells (Tables 5, 4); it must operate as a spillover to non-AI patents (Table 4); and it must be concentrated among patent-intensive incumbents (Table 6). The Schumpeterian quality-ladder model in the next section delivers all three through a single channel: a shift in the step-size distribution toward small steps raises incumbent franchise value, raises internal R&D, and lowers the creative-destruction rate that incumbents tolerate.

[2] A fully causal firm-size design would require a firm–CPC4 employment link, which is not available in the processed data.

3 Theoretical Framework

3.1 Environment

A representative household has linear preferences

$$U = \int_0^\infty e^{-\rho t} C(t)\, dt. \tag{3}$$

We adopt linear utility, which anchors the equilibrium interest rate at $r = \rho$. This severs the mechanical feedback from endogenous growth to the interest rate, allowing us to isolate the AI step-size shock on R&D incentives in closed form.

The final good is produced under perfect competition by

$$Y = \frac{L^\beta}{1-\beta} \int_0^1 q_j^\beta k_j^{1-\beta}\, dj. \tag{4}$$

Each intermediate line j is run by a single monopolist with marginal cost $1 - \beta$. Standard monopoly pricing, derived in Section 1 of the supplementary derivations, yields machine demand $k_j^* = q_j L$ and flow profit $\pi_j = \beta q_j L$. Define the per-unit-quality coefficient

$$\pi \equiv \beta L. \tag{5}$$

Crucially, π is not a free parameter but a function of the labour share β and the population L.

3.2 Step-size distribution

External innovations draw a step size $s \in \{s_L, s_H\}$ with $0 < s_L < s_H$ and

$$\Pr(s = s_L \mid \theta) = p(\theta) \in (0, 1). \tag{6}$$

Assumption 1 (A1: AI shifts mass toward small steps)**.** $p : \mathbb{R}_{++} \to (0,1)$ *is differentiable with* $p'(\theta) > 0$*.*

The expected external step size and its derivative are

$$\bar{s}(\theta) = p(\theta)\, s_L + (1 - p(\theta))\, s_H, \tag{7}$$

$$\frac{d\bar{s}}{d\theta} = p'(\theta)\,(s_L - s_H) \;<\; 0. \tag{8}$$

3.3 Production, profits, and the Arrow replacement effect

Assumption 2 (A2: Arrow replacement)**.** *Incumbents' optimal external R&D effort on lines they already own is zero.*

Proof of A2. Conjecture $V(\mathbf{q}) = A\sum_j q_j$. An entrant succeeding on line j with step s gains $A(q_j + s\bar{Q})$. An incumbent already owning line j gains the same post-innovation value but loses the status-quo Aq_j, so its marginal benefit of external R&D on its own line is below the entrant's by exactly $Aq_j > 0$. Since the entrant's marginal benefit equals marginal cost at the free-entry condition, the incumbent's marginal benefit is strictly below cost, and the optimum is the corner $x^{\text{inc},*} = 0$. □

3.4 Internal R&D (incumbents)

The incumbent chooses internal-improvement intensity z at flow cost $\chi z^{\psi}\bar{q}$; success arrives at rate z and raises line quality by λq_j. With the Arrow corner, the per-line HJB is

$$\rho A \;=\; \pi - \chi z^{\psi} + zA\lambda - \tau A, \tag{9}$$

where τ is the equilibrium creative-destruction rate.

3.5 External R&D (entrants only)

A free entrant chooses arrival rate x_e at cost $\chi x_e^{\psi} \bar{q}$. Successful innovation yields expected value $A\bar{q}(1 + \bar{s}(\theta))$. Free entry equates marginal benefit and marginal cost:

$$A\,(1 + \bar{s}(\theta)) \;=\; \chi\,\psi\,x^{\psi-1}. \tag{10}$$

Defining the entry-cost parameter $\nu \equiv \chi\psi x^{\psi-1}$, the free-entry condition rearranges to

$$A(\theta) \;=\; \frac{\nu}{1 + \bar{s}(\theta)}. \tag{11}$$

3.6 Equilibrium definition

Definition 1 (Equilibrium)**.** *An equilibrium is a tuple $(A(\theta), z(\theta), x, \tau(\theta), g(\theta))$ satisfying:*

(i) free-entry: $A(1 + \bar{s}(\theta)) = \chi\psi x^{\psi-1}$;

(ii) incumbent internal-R&D FOC: $\chi\psi z^{\psi-1} = A\lambda$;

(iii) external-R&D invariance: $x = (\nu/(\chi\psi))^{1/(\psi-1)}$;

(iv) HJB balance: $\tau = \pi/A + z\lambda(1 - 1/\psi) - \rho$;

(v) $\bar{Q}$ law of motion: $g = z\lambda + \tau\bar{s}$.

3.7 BGP characterisation

Definition 2 (Balanced growth path)**.** *A BGP is an equilibrium in which A, z, x, τ, g are constant in time and $\bar{Q}(t)$ grows at rate $g(\theta)$.*

Proposition 1 (Existence and uniqueness)**.** *Under Assumptions 1–2 and the transversality condition $\rho > g(\theta)$, a unique BGP exists. The equilibrium tuple is determined recursively in the order $x \to A(\theta) \to z(\theta) \to \tau(\theta) \to g(\theta)$, and the BGP growth rate is*

$$g(\theta) \;=\; z(\theta)\lambda + \tau(\theta)\,\bar{s}(\theta). \tag{12}$$

Proof. Step (iii) of Definition 1 pins down x as a function of primitives only. Equation (11) then determines $A(\theta)$, and step (ii) determines $z(\theta)$ as a function of $A(\theta)$. The HJB balance (iv) then determines $\tau(\theta)$, and (v) closes the system. Each step uses only previously determined objects; uniqueness is immediate. See Section 9 of the supplementary derivations. □

4 Theoretical Results

Proposition 2 (Mean Novelty)**.** *As θ rises, the expected external step size strictly decreases: $d\bar{s}/d\theta = p'(\theta)\,(s_L - s_H) < 0$.*

Proof. Direct from (8), A1, and $s_L < s_H$. □

Interpretation. A rise in AI exposure is modelled as a rise in the probability $p(\theta)$ that any given external innovation is of the small-step type. Because the step-size distribution is a two-point mixture, the mean shifts mechanically toward s_L. This single primitive captures the empirical observation that AI-exposed CPC4 classes produce more derivative inventions on average.

Proposition 3 (Variance)**.** $\mathrm{Var}(s \mid \theta)$ *strictly increases in θ if and only if $p(\theta) < 1/2$: $d\mathrm{Var}/d\theta = (1 - 2p(\theta))\,p'(\theta)\,(s_H - s_L)^2$.*

Proof. Direct from $\mathrm{Var}(s \mid \theta) = p(\theta)(1 - p(\theta))(s_H - s_L)^2$ and A1. □

Interpretation. The cross-sectional spread of novelty rises as long as the small-step type is not yet a majority of innovations. Combined with Proposition 2, the model predicts the asymmetric distributional pattern documented in Table 5: the mean falls and the lower tail falls by more than the upper tail rises.

Proposition 4 (Franchise value)**.** *$A(\theta)$ strictly increases in θ: $dA/d\theta = -\nu/(1+\bar{s})^2 \cdot d\bar{s}/d\theta > 0$.*

Proof. Differentiate (11) and apply Proposition 2. □

Interpretation. Smaller expected steps mean any successor innovation is a smaller threat to the value of an existing line. The free-entry condition forces the marginal-benefit-to-marginal-cost ratio of an entrant back to one, but the marginal benefit per unit of $\bar{q}$ is now larger. The franchise value A that a single monopoly line commands therefore rises.

Proposition 5 (Internal R&D)**.** *$z(\theta)$ strictly increases in θ. From the FOC $z = (A\lambda/(\chi\psi))^{1/(\psi-1)}$ and Proposition 4.*

Proof. z is strictly increasing in A for $\psi > 1$; combine with Proposition 4. □

Interpretation. Higher franchise value raises the marginal return to internal R&D, so incumbents intensify within-line improvements. The model predicts a within-firm productivity boost from AI, consistent with much of the firm-level empirical literature, but it is the wrong margin for aggregate growth.

Proposition 6 (External R&D invariance)**.** *x is invariant to θ under the free-entry zero-profit condition: $x = (\nu/(\chi\psi))^{1/(\psi-1)}$ contains no θ.*

Proof. Direct from (10). □

Interpretation. Because the free-entry condition is the only force that determines x on a BGP, the entry rate is invariant in θ. All adjustment to the AI shock works through A, z, and τ; no force in the model induces additional entry as θ rises.

Proposition 7 (Creative destruction)**.** *Under Assumption A2 (i.e. $\pi > \chi z^{\psi}$ in the transversality region), τ strictly decreases in θ: $d\tau/dA = (-\pi + \chi z^{\psi})/A^2 < 0$.*

Proof. From (14) below and the FOC, differentiate τ in A. Assumption A2 implies the numerator is negative; combine with Proposition 4. □

Interpretation. Higher franchise value raises the implicit cost to incumbents of being displaced; in equilibrium, the destruction rate adjusts down to maintain HJB balance. Slower creative destruction is the second pillar of the AI Growth Paradox.

Proposition 8 (AI Growth Paradox)**.** *$dg/d\theta = \lambda\, dz/d\theta + \bar{s}\, d\tau/d\theta + \tau\, d\bar{s}/d\theta$ has ambiguous sign. A sufficient condition for $dg/d\theta < 0$ is*

$$\bar{s}\,\big|d\tau/d\theta\big| + \tau\,\big|d\bar{s}/d\theta\big| \;>\; \lambda\, dz/d\theta. \tag{13}$$

Proof. Direct differentiation of $g = z\lambda + \tau\bar{s}$. □

Interpretation. Three forces act on aggregate growth. The within-firm channel ($\lambda\, dz/d\theta > 0$) raises growth as incumbents intensify internal R&D. The destruction channel ($\bar{s}\, d\tau/d\theta < 0$) lowers growth because slower displacement reduces the rate at which large external steps arrive. The composition channel ($\tau\, d\bar{s}/d\theta < 0$) lowers growth because the average external step is smaller. When the two negative channels dominate the positive channel—inequality (13)—AI lowers aggregate growth. The current draft treats this as a theoretical possibility, not as a structurally estimated quantitative result.

Proposition 9 (Dynamic misallocation). *$x_{\mathrm{SP}}(\theta) > x$ and $d(x_{\mathrm{SP}} - x)/d\theta > 0$. The standing-on-shoulders externality makes the planner's shadow value μ strictly exceed A, with the wedge μ/A increasing in θ: $x_{\mathrm{SP}}/x = (\mu/A)^{1/(\psi-1)}$.*

Proof. Compare the planner's FOC $\mu(1+\bar{s}) = \chi\psi x_{\mathrm{SP}}^{\psi-1}$ with the decentralised FEC. The planner internalises the standing-on-shoulders externality whereby each successful external innovation permanently raises $\bar{Q}$ and hence the size $s\bar{Q}$ of every future external step; private entrants do not. □

Interpretation. The decentralised entry rate is too low at every θ, and the gap widens as AI raises the franchise value. AI therefore not only slows growth in equilibrium but also widens the gap between equilibrium and planner growth.

Proposition 10 (Scarce labour). *With $L_R = \chi z^{\psi} + \chi x^{\psi}$ fixed and the FEC dropped, the ratio of FOCs gives*

$$\frac{\chi\psi z^{\psi-1}}{\chi\psi x^{\psi-1}} = \frac{\lambda}{1+\bar{s}(\theta)}.$$

The right-hand side rises in θ, so z rises and x falls.

Proof. See Section 14 of the supplementary derivations. □

Interpretation. When R&D labour is scarce rather than infinitely supplied, the entrenchment mechanism is reinforced: incumbents bid R&D labour away from entrants. The Paradox condition is then strictly easier to satisfy.

5 Robustness of the Theoretical Mechanism

5.1 Scarce R&D labour

Proposition 10 shows that fixing the R&D labour pool and dropping the FEC strengthens the entrenchment effect. The qualitative comparative statics on A, z, and τ are unchanged; the magnitude of the destruction-rate decline is amplified.

5.2 CRRA preferences

Replacing linear utility (3) with the CRRA aggregator $C^{1-\sigma}/(1-\sigma)$ adds the Euler equation $r = \rho + \sigma g(\theta)$ to the system. The endogenous interest rate provides additional dampening: as g falls, r falls, which raises A further. This second-round effect can reinforce the franchise-value channel, but its quantitative importance is left to a future calibrated model.

6 Conclusion

We have constructed a semantic novelty index for the universe of US patents from 2006 to 2023, documented an asymmetric and tail-biased relationship between AI exposure and the distribution of innovation, externally benchmarked the novelty distribution against citations, KPSS values, maintenance-fee outcomes, and Compustat firm-level fundamentals, established that the lower-tail compression is a true *spillover* from rising AI intensity to surrounding non-AI inventions, and embedded the empirical mechanism in a Schumpeterian model whose nine propositions are SymPy-verified. The evidence supports the claim that AI exposure compresses the lower tail of inventive distance for both AI and non-AI patents within an exposed CPC4 class, and that this compression operates through the entrenchment channel formalised in the model.

Our findings point to two implications for innovation policy. First, external-R&D subsidies targeted at entrants are the natural corrective instrument in the model because private entrants do not internalise the standing-on-shoulders externality of Proposition 9. Second, antitrust policy should attend to AI-enabled increases in incumbent franchise value, because the same channel that raises incumbent rents can reduce creative destruction. Quantifying optimal subsidy and enforcement magnitudes is a natural next step.

Several extensions suggest themselves. The semantic novelty measure is built from patent abstracts; extending it to non-patented AI-enabled process improvements — for instance using firm-level text from earnings calls or patent claims — would broaden coverage. Allowing the step-size mixture $p(\theta)$ to vary across CPC4 classes would tighten the mapping from cross-class heterogeneity to aggregate growth. A heterogeneous-firm extension with separate entrant and incumbent R&D accounting, in the spirit of Akcigit and Kerr (2018), would permit a fully identified version of the calibration in Appendix B and would let the planner target incumbents and entrants separately.

Statements and Declarations

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Data Availability. All raw data used in this study are publicly available. USPTO patent metadata and citations are from PatentsView (`https://patentsview.org`); the AI-patent flag is from the USPTO Office of the Chief Economist's Artificial Intelligence Patent Dataset (Giczy and Pairolero, 2022). The KPSS patent-value file is the 2024 release of Kogan et al. (2017), available at `https://github.com/KPSS2017/Update`. USPTO maintenance-fee events are from the USPTO Bulk Data Storage System. Compustat fundamentals annual and the CRSP/Compustat Merged link table are accessed via Wharton Research Data Services (WRDS); these data are licensed and cannot be redistributed but are accessible to subscribing institutions. Macro moments are from FRED (TFP series `RTFPNAUSA632NRUG`), the U.S. Census Bureau Business Dynamics Statistics, and NSF/NCSES BERD Tables 2 and 30. All processed datasets, code to reproduce the empirical pipeline, and SymPy verification of the model propositions are available on the author's GitHub repository upon publication; a replication snapshot is included as Online Resource 1.

Author Contributions. Sole-authored work. The author conceived the project, conducted the empirical analysis, derived the theoretical results, and prepared the manuscript.

Use of AI tools. The author used AI assistants to support code development, table formatting, and copy editing. All analytical decisions, derivations, regression specifications, and interpretations are the author's own; the author takes full responsibility for the final content.

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A Proofs

The τ expression used in Proposition 7 follows from substituting the internal-R&D FOC $\chi z^{\psi} = A\lambda z/\psi$ into (9):

$$\tau(\theta) \;=\; \frac{\pi}{A(\theta)} + z(\theta)\lambda\left(1 - \frac{1}{\psi}\right) - \rho. \tag{14}$$

Section 8 of the supplementary derivations contains the full SymPy verification of all nine propositions, in which each proposition's symbolic check returns true.

B Quantitative Implications of the Model

The natural next step is to discipline the BGP with patent-panel and macro moments. The new external datasets sharpen the relevant targets. BDS data imply a pre-period entry-rate proxy of 10.54 percent and a post-period proxy of 10.47 percent (establishment-entry rate, the available proxy in the public BDS file). NSF/NCSES BERD Table 30 implies R&D-to-sales intensity of 14.27 percent among firms with fewer than 50 domestic employees and 3.83 percent among firms with at least 5,000 domestic employees (Table 7). These moments discipline the entrant and incumbent effort margins that the original patent moments could not separately identify.

Table 7: External macro moments for quantitative calibration.

Moment	Pre / small	Post / large
Entry rate (BDS proxy)	10.54%	10.47%
Firm death rate (BDS)	9.05%	8.14%
R&D/sales intensity (BERD)	14.27%	3.83%

Table 8: Rejected SMM diagnostic. Bootstrap standard errors in parentheses; 95% percentile intervals in brackets. $B = 500$ converged of $B = 500$ requested. These estimates are reported to document non-identification and are not used for quantitative claims.

Parameter	Description	Estimate	95% CI
$\widehat{chi\_hat}$	internal R&D cost scale	10.0000 (0.0000)	[10.0000, 10.0000]
$\widehat{chi\_tilde}$	external R&D cost scale	10.0000 (0.0000)	[10.0000, 10.0000]
$\widehat{s\_L}$	low step size	0.0100 (0.0000)	[0.0100, 0.0100]
$\widehat{s\_H}$	high step size	0.5000 (0.0000)	[0.5000, 0.5000]
$\widehat{p\_0}$	prob of low step at theta=0	0.8000 (0.0000)	[0.8000, 0.8000]
$\widehat{p\_1}$	sensitivity of p to theta	0.3397 (0.0444)	[0.2586, 0.4249]
$\widehat{phi\_0}$	novelty index intercept	0.6366 (0.0063)	[0.6235, 0.6485]
$\widehat{phi\_1}$	novelty index slope	0.3554 (0.0081)	[0.3393, 0.3707]
J-statistic = 1897.588 (df = 1, $p = 0.000$)			

We attempted a two-step SMM using pre- and post-period mean novelty, tenth-percentile novelty, and the 90–10 spread, together with pre-period TFP growth from FRED and aggregate R&D intensity from NSF.

The exercise is informative about the *direction* of the step-size shift but not about its joint level with the other seven parameters. Five of the eight parameters lie at imposed bounds, the overidentification statistic is $J = 1897.6$ on one degree of freedom, and the implied growth rate violates the transversality condition of Proposition 1. We therefore use the calibration to discipline the qualitative direction of $p(\theta)$ rather than as a joint structural estimate; a fully identified version of this exercise would require a heterogeneous-firm extension with separate entrant and incumbent R&D

Table 9: Empirical vs. model-implied moments in the rejected SMM run.

Moment	Empirical (SE)	Model	Diff	% Diff
novelty_mean_pre	0.6768 (0.0008)	0.6687	-0.0081	-1.2%
novelty_mean_post	0.6709 (0.0009)	0.6634	-0.0075	-1.1%
novelty_p10_pre	0.6027 (0.0009)	0.5861	-0.0166	-2.8%
novelty_p10_post	0.5948 (0.0010)	0.5876	-0.0072	-1.2%
novelty_spread_pre	0.1550 (0.0004)	0.1651	+0.0102	+6.6%
novelty_spread_post	0.1589 (0.0004)	0.1517	-0.0072	-4.5%
tfp_growth_pre	0.0095 (0.0030)	0.0311	+0.0216	+227.6%
rd_intensity_pre	0.0270 (0.0010)	0.0710	+0.0440	+162.9%

accounting, in the spirit of Akcigit and Kerr (2018).

C Data

The patent universe consists of USPTO utility patents granted between 2006 and 2023, drawn from PatentsView; the artificial-intelligence flag is from the USPTO Office of the Chief Economist's Artificial Intelligence Patent Dataset (Giczy and Pairolero, 2022). Abstract embeddings are computed using the `all-MiniLM-L6-v2` sentence-transformer model (Reimers and Gurevych, 2019). Patent-level novelty is the cosine-distance index defined in Section 2.

External patent-level validation data are: five-year forward citations constructed from PatentsView patent-citation pairs; market-based patent values from the 2024 release of the Kogan et al. (2017) KPSS file; and eight-year maintenance-fee expirations parsed from the USPTO Maintenance Fee Events file. Firm-level validation data are Compustat fundamentals annual and the CRSP/Compustat Merged link table, both accessed through Wharton Research Data Services. Patents are bridged to firms using the KPSS patent–permno–permco match table; the resulting gvkey–year panel covers 20,291 firm-years across 3,424 public firms over 2006–2023.

The non-AI spillover panel is constructed by reaggregating patent-level novelty over non-AI patents only and merging back to the same CPC4–year cells used in the headline regressions. Macro moments used in Appendix B are: TFP growth from FRED (series `RTFPNAUSA632NRUG`); firm-dynamics moments from the U.S. Census Bureau Business Dynamics Statistics 2023 release; and R&D-intensity moments from NSF/NCSES BERD Tables 2 and 30. Sample restrictions throughout are: patents with non-empty abstracts, at least one CPC4 assignment, and at least one disambiguated assignee.

D Additional Empirical Results

Descriptive pre/post moments and density

For descriptive context, mean novelty in AI-exposed CPC4 classes falls from 0.6768 in the pre-period (2006–2011) to 0.6709 in the post-period (2019–2023), a decline of 0.0059 index points. The tenth-percentile novelty falls from 0.6027 to 0.5948 (a decline of 0.0079), and the 90–10 spread widens from 0.1550 to 0.1589.

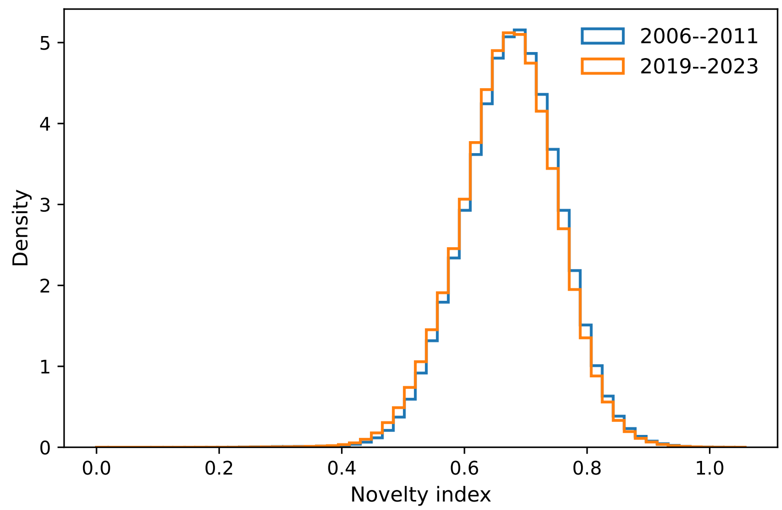

Figure 1: Kernel density of CPC4-year mean novelty, AI-exposed classes, pre (2006–2011) versus post (2019–2023).